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$$\begin{aligned} & \int_{-1}^2 x \lfloor x \rfloor dx \\ &= \int_{-1}^0 x \lfloor x \rfloor dx + \int_0^1 x \lfloor x \rfloor dx + \int_1^2 x \lfloor x \rfloor dx \\ &= \int_{-1}^0 x \cdot (-1) dx + \int_0^1 x \cdot 0 dx + \int_1^2 x \cdot 1 dx \\ &= - \int_{-1}^0 x dx + 0 + \int_1^2 x dx \\ &= - \left[\frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_1^2 \\ &= - \left(0 - \frac{1}{2} \right) + \left(2 - \frac{1}{2} \right) \\ &= \frac{1}{2} + \frac{3}{2} \\ &= 2 \end{aligned}$$

References